

Math 133 Written Homework 3 Solutions

0. **Assignment record.** Responses depend on the student's collaborators, time spent, and use of generative AI.

1. **An investment model.**

- (a) The initial investment is \$2,000. The factor 1.10 means the investment grows by 10% per year. After three years,

$$A(3) = 2000(1.10)^3 = \boxed{\$2,662.00}.$$

- (b) Dividing by 2000 and taking logarithms gives

$$2000(1.10)^t = 4000 \implies (1.10)^t = 2 \implies t \ln(1.10) = \ln 2.$$

Thus $t = \ln 2 / \ln(1.10) \approx 7.27254$, so the doubling time is $\boxed{7.3 \text{ years}}$. The first whole-number year at which the value reaches at least \$4,000 is year $\boxed{8}$. Indeed, $A(7) \approx \$3,897.43$ and $A(8) \approx \$4,287.18$.

2. **Total cost, average cost, and the next item.**

- (a) The additional costs are

$$C(11) - C(10) = 239 - 230 = \$9, \quad C(12) - C(11) = 249 - 239 = \$10.$$

The 11th item adds \$9 to total cost, and the 12th adds \$10.

- (b) The average rates of change are

$$\frac{230 - 150}{10 - 0} = 8, \quad \frac{345 - 230}{20 - 10} = 11.50, \quad \frac{480 - 345}{30 - 20} = 13.50$$

in dollars per item, respectively. Total cost increases fastest on average over $[20, 30]$.

- (c) $C(10)/10 = 230/10 = \$23$ per item averages the entire cost of the first ten items, including fixed costs, over those ten items. In contrast, $C(11) - C(10) = \$9$ is only the additional cost of making the next item. These measure different quantities and need not agree.

3. From a difference quotient to a marginal estimate.

(a) Substitution gives

$$C(40) = 100 + 4(40) + 0.10(40)^2 = 420,$$

$$C(41) = 100 + 4(41) + 0.10(41)^2 = 432.10.$$

Therefore the exact additional cost of the 41st item is

$$C(41) - C(40) = 432.10 - 420 = \boxed{\$12.10}.$$

(b) We have $C(45) = 100 + 180 + 202.50 = 482.50$, so

$$\frac{C(45) - C(40)}{5} = \frac{482.50 - 420}{5} = \boxed{12.50 \text{ dollars per item}}.$$

This is the average increase in total cost per item when production rises from 40 to 45 items. It averages the additional costs of five items, while part (a) concerns only the 41st item. The additional costs increase with production, so these answers differ.

(c) The secant slope is the change in cost divided by the change in quantity:

$$m_{\text{sec}} = \frac{C(40+h) - C(40)}{(40+h) - 40} = \frac{100 + 4(40+h) + 0.10(40+h)^2 - 420}{h}.$$

Since $C(40+h) = 420 + 12h + 0.10h^2$, for $h \neq 0$ this simplifies to

$$m_{\text{sec}} = \frac{12h + 0.10h^2}{h} = \boxed{12 + 0.10h} \quad \text{dollars per item.}$$

The model requires $0 \leq 40+h \leq 100$, so $-40 \leq h \leq 60$, with $h \neq 0$. When $h = 0$ the two points coincide, so the secant slope fraction is undefined, even though the simplified expression can be evaluated there.

(d) The quotient values are

h (items)	Difference quotient (dollars per item)
1	12.100
0.1	12.010
0.01	12.001

The small-increment values suggest a marginal cost rate of about \$12 per item at $q = 40$. This gives an estimated cost of $12(1) = \$12.00$ for the 41st item. The estimate is \$0.10 below the exact additional cost, \$12.10. The marginal rate describes change near 40 items; it does not capture all of the rise in the cost rate over the full additional item.

4. Interpreting marginal rates.

- (a) $P(100) = 1600 - 1800 = -\200 : the business has a total loss of \$200 at 100 items. The marginal profit rate is $11 - 8 = \$3$ per item, so producing and selling one more item is expected to increase profit by about \$3.
- (b) The one-item estimates are

$$C(101) \approx 1800 + 8 = \$1,808, \quad R(101) \approx 1600 + 11 = \$1,611,$$

$$P(101) \approx 1611 - 1808 = -\$197.$$

If the rates remain approximately constant for the next five items,

$$P(105) \approx P(100) + 5(3) = -200 + 15 = \boxed{-\$185}.$$

- (c) No. The next item's estimated additional revenue, \$11, exceeds its additional cost, \$8, so it reduces the loss by about \$3. Negative total profit means the business is currently losing money; positive marginal profit means that increasing production slightly improves that profit. The business can still have a loss after an improvement.

5. Does one more sale help?

- (a) Subtracting costs from revenue gives

$$P(q) = 40q - 0.25q^2 - (100 + 10q + 0.05q^2) = \boxed{-0.30q^2 + 30q - 100}.$$

- (b) Using $(q + 1)^2 - q^2 = 2q + 1$, the one-item changes simplify to

$$\Delta R = 39.75 - 0.50q, \quad \Delta C = 10.05 + 0.10q, \quad \Delta P = 29.70 - 0.60q.$$

Thus, with all changes in dollars,

Starting q	$R(q + 1) - R(q)$	$C(q + 1) - C(q)$	$P(q + 1) - P(q)$
20	29.75	12.05	17.70
60	9.75	16.05	-6.30

In each row $\Delta P = \Delta R - \Delta C$: $29.75 - 12.05 = 17.70$ and $9.75 - 16.05 = -6.30$.

- (c) At $q = 20$, one more item increases profit by \$17.70 because additional revenue exceeds additional cost. At $q = 60$, one more item decreases profit by \$6.30 because additional cost exceeds additional revenue.
- (d) Total profit falls by \$100 at every production level: $\tilde{P}(q) = P(q) - 100$. However,

$$\tilde{P}(q + 1) - \tilde{P}(q) = [P(q + 1) - 100] - [P(q) - 100] = P(q + 1) - P(q).$$

The constant cancels, so both one-item profit changes remain unchanged.

6. Excel investigation: marginal changes and a production decision.

- (a) Enter $q = 0, 1, \dots, 100$ in A2:A102. The initial formulas are

Cell	Formula	Cell	Formula
B2	=2000*A2/(A2+20)	E2	=B3-B2
C2	=200+8*A2+0.04*A2^2	F2	=C3-C2
D2	=B2-C2	G2	=D3-D2

Fill B2:D2 through row 102, and E2:G2 through row 101. E102:G102 remain blank. The selected rows below contain the requested values; all monetary entries are displayed to two decimal places.

q	$R(q)$	$C(q)$	$P(q)$	ΔR	ΔC	ΔP
0	0.00	200.00	-200.00	95.24	8.04	87.20
1	95.24	208.04	-112.80	86.58	8.12	78.46
2	181.82	216.16	-34.34	79.05	8.20	70.85
20	1000.00	376.00	624.00	24.39	9.64	14.75
38	1310.34	561.76	748.58	11.69	11.08	0.61
39	1322.03	572.84	749.19	11.30	11.16	0.14
40	1333.33	584.00	749.33	10.93	11.24	-0.31
41	1344.26	595.24	749.02	10.58	11.32	-0.74
42	1354.84	606.56	748.28	10.24	11.40	-1.16
60	1500.00	824.00	676.00	6.17	12.84	-6.67
100	1666.67	1400.00	266.67	—	—	—

- (b) At $q = 20$, the 21st item adds approximately \$24.39 in revenue and \$9.64 in cost, increasing profit by \$14.75. At $q = 60$, the 61st item adds about \$6.17 in revenue but \$12.84 in cost, reducing profit by \$6.67.

At $q = 0$, $E2 - F2$ is approximately $95.238095 - 8.04 = 87.198095$, matching $G2$. For a negative-profit-change row, $q = 60$ gives $6.172840 - 12.84 = -6.667160$, matching $G62$. These identities hold before rounding; subtracting rounded displays can occasionally differ by a cent.

- (c) The two graphs on the next page plot the tabulated values with connected markers. Profit peaks near 40 items. The additional revenue per item decreases, while the additional cost per item increases.
- (d) The largest profit among whole-number quantities $0 \leq q \leq 100$ occurs at $q = 40$:

$$P(40) = \frac{2000(40)}{60} - (200 + 8(40) + 0.04(40)^2) = \$749.33 \text{ (rounded).}$$

Immediately before and after this quantity,

$$P(40) - P(39) \approx \$0.14 > 0, \quad P(41) - P(40) \approx -\$0.31 < 0.$$

The table shows $\Delta R > \Delta C$ for $q = 0, \dots, 39$ and $\Delta R < \Delta C$ for $q = 40, \dots, 99$. Thus profit increases up to 40 items and decreases thereafter.

Problem 6: graphs of the tabulated values.

These graphs use the same values and connected-marker format as the requested Excel scatterplots. The second graph's input q denotes the starting quantity; its changes are for item $q + 1$.

Positive marginal profit describes an *increase* in profit; positive total profit means revenue exceeds total cost. For example, $P(0) = -\$200$ while ΔP at $q = 0$ is positive. Conversely, $P(60) = \$676 > 0$ while its one-item profit change is negative.

